ECommerce Sales Analysis

Project Objective:

Analyse overall sales performance using total sales, quantity sold, and transaction-level data.Identify top-performing products and categories based on sales revenue and quantity sold.Evaluate customer purchasing behavior by analyzing customer demographics such as age, gender, city, and loyalty level.Compare store performance across different regions, cities, and store types.Analyze sales trends over time using order dates to identify monthly, quarterly, or yearly patterns.Evaluate the impact of discounts on sales revenue and customer purchases.Analyze payment methods to understand customer preferences and transaction patterns.Identify high-value customers and products that contribute significantly to overall revenue.Create Pivot Tables and dashboards to summarize the data and present key business insights visually.

Dataset Description:

E-Commerce Sales Dataset contains transactional sales information from an e-commerce business. The dataset captures details about sales transactions, customers, products, stores, payment methods, quantities, prices, discounts, and total sales amounts. It also includes customer attributes such as gender and loyalty level, product details such as category, sub-category and brand, and store details such as store type and region.

The dataset can be used to analyze **sales performance, category performance, customer behavior, regional performance, payment methods, discounts, and yearly sales trends**. It supports Descriptive, Diagnostic, Predictive, and Prescriptive Analytics using Excel Pivot Tables and charts.

Column Description

Column Name	Description
Sales_ID	Unique identification number for each sales transaction.
Order_Date	Date on which the customer placed the order
Customer_ID	Unique identification number of the customer.
Product_ID	Unique identification number of the product sold.
Store_ID	Unique identification number of the store where the transaction was recorded
Quantity	Number of units purchased in the transaction.
Unit_Price	Selling price of one unit of the product.
Discount	Discount percentage applied to the transaction.
Payment_Type	Payment method used by the customer, such as Credit Card, PayPal, or COD.
Total_Amount	Final sales amount after considering quantity, price, and discount.
Customer_Name	Name of the customer.
Gender	Gender of the customer.
Age	Age of the customer.
Loyalty_Level	Customer loyalty category based on their relationship or purchase behavior.
Product_Name	Name of the product.
Category	Main product category, such as Sports, Electronics, Beauty, Clothing, or Home.
Brand	Brand associated with the product.
Cost	Cost incurred by the business to obtain or produce the product.
Stock	Available quantity of the product in inventory.
Store_Name	Name or identifier of the store.
Store_Type	Type of store, such as Flagship, Online, or Outlet.
Region	Geographic region where the store operates.
City	City/location of the store.

Data cleaning & transformation:
Formulas:



Name	Age	Gender	City	State	Country	Loyalty_Level	Customer ID Standardization
Sheena Bonilla	55	Male	Donaldton	North Carolina	USA	Unknown	
Eugene Nelson	26	Female	Phillipsmouth	North Dakota	USA	Platinum	In the Customer ID column, inconsistent values such as eg:"C-1464" were identified.
Dr. Benjamin Price	29	Female	Kylebury	Alabama	USA	Platinum	The values were standardized to the correct format "Cust1464" using the **Replace Values** option.
Jeremy Harris	59	Female	Thomastown	Ohio	USA	Bronze	
Caroline Collins	64	Female	Nicholasshire	California	USA	Silver	**Name Cleaning and Standardization**
Christine Vaughn	24	Male	East Amandamouth	Oregon	USA	Bronze	
Amber Jackson	29	Female	Deborahborough	South Dakota	USA	Platinum	* The Name column was cleaned to remove unnecessary spaces and inconsistent formatting.
Daniel Boyd	29	Female	West Pamelafurt	West Virginia	USA	Bronze	* **Trim** was used to remove extra spaces.
Melissa Howard	53	Female	Maynardton	New Hampshire	USA	Gold	* **Clean** was applied to remove unwanted/non-printable characters.
Ms. Maria Leach	64	Male	Melissashire	Pennsylvania	USA	Platinum	* **Prope**r Case was used to standardize names with the correct capitalization.
Dominic Page	53	Male	Port Michaelburgh	Pennsylvania	USA	Bronze	**proper(trim(clean(b2)))**
Brian Lopez	28	Female	East Krista	New York	USA	Gold	
Kenneth Lambert	55	Female	North Kathryn	Kansas	USA	Platinum	**City Name Standardization**
Nathan Hernandez	32	Female	New Becky	Montana	USA	Silver	
Ashley Cohen	53	Female	Lake Kristamouth	Nebraska	USA	Silver	* The City Name column was cleaned and standardized.
Laura Alvarez	49	Female	Lawsonhaven	Washington	USA	Platinum	* Extra spaces and unwanted characters were removed using Trim and Clean.
John Scott	24	Female	South Christopher	Pennsylvania	USA	Bronze	* Proper Case was applied to maintain a consistent city-name format.
Adam Cruz	60	Female	Brandonville	Georgia	USA	Platinum	**proper(trim(clean(e2)))**
Rhonda Moss	43	Male	South William	Kansas	USA	Unknown	
Michael Torres	46	Female	Gonzalezfort	New Jersey	USA	Platinum	
Bridget Allen	59	Male	Lake Shaneburgh	Delaware	USA	Silver	**Country Name Standardization**
Jeff Martinez	36	Female	Smithburgh	Oklahoma	USA	Gold	
Paul Fox	33	Female	Michaelton	Hawaii	USA	Silver	* In the Country column, the value **"United States of America"** was replaced with **"USA"**.
Jonathan Steele	52	Female	Molinabury	Oregon	USA	Silver	* The **Replace Values** option was used to maintain a short and consistent country format.
Heather Collins	40	Male	Rogerschester	Idaho	USA	Platinum	
Diana Taylor	59	Female	Vasqueztown	California	USA	Bronze	**Loyalty Level – Missing Value Handling**
Amber Carlson	18	Female	East Michellemouth	Missouri	USA	Gold	
Leslie Johnson	31	Male	Mollyshire	Utah	USA	Unknown	* Blank or missing values in the Loyalty Level column were identified.
Joshua Ryan	27	Female	New David	New Jersey	USA	Platinum	* The missing values were replaced with **"Unknown"** to clearly indicate unavailable information.
David Arnold	54	Female	East Jennifer	California	USA	Silver	* This helped prevent **blank values** from affecting further analysis.
Jessica Acevedo	64	Female	Amybury	Pennsylvania	USA	Bronze	
Sarah Vance	41	Male	Johnsonville	Oregon	USA	Silver	

Customer ID Standardization

In the Customer ID column, inconsistent values such as eg:"C-1464" were identified.
The values were standardized to the correct format "Cust1464" using the **Replace Values** option.

Name Cleaning and Standardization

* The Name column was cleaned to remove unnecessary spaces and inconsistent formatting.
* **Trim** was used to remove extra spaces.
* **Clean** was applied to remove unwanted/non-printable characters.
* **Prope**r Case was used to standardize names with the correct capitalization.

proper(trim(clean(b2)))

City Name Standardization

* The City Name column was cleaned and standardized.
* Extra spaces and unwanted characters were removed using Trim and Clean.
* Proper Case was applied to maintain a consistent city-name format.

proper(trim(clean(e2)))

Country Name Standardization

* In the Country column, the value "**United States of America"** was replaced with "**USA**".
* The **Replace Values** option was used to maintain a short and consistent country format.

Loyalty Level – Missing Value Handling

* Blank or missing values in the Loyalty Level column were identified.
* The missing values were replaced with "**Unknown**" to clearly indicate unavailable information.
* This helped prevent **blank values** from affecting further analysis.



Product_ID	Product_Brand	Category	Sub_Category	Cost	Stock
PROD0	Brother Laser Printer - Hunt Inc	Electronics	Computer	₹ 112.42	416.00
PROD1	Men'S Graphic T-Shirt - Terry-Moore	Clothing	Tops	₹ 342.56	181.00
PROD2	Women'S Casual Hoodie - Sims Inc	Clothing	Outerwear	₹ 254.62	352.00
PROD3	Men'S Running Jacket - Pineda Ltd	Clothing	Outerwear	₹ 364.67	452.00
PROD4	Women'S Yoga Pants - Gonzales-Davis	Clothing	Bottoms	₹ 306.46	487.00
PROD5	Men'S Leather Belt - Baldwin, Garza And Gonzalez	Clothing	Accessories	₹ 261.00	59.00
PROD6	Herbal Face Moisturizer - Davidson Plc	Beauty	Skincare	₹ 312.80	350.00
PROD7	Led Table Lamp - Harrell Llc	Home	Lighting	₹ 218.86	170.00
PROD8	Wireless Bluetooth Earbuds - Perez-Johnston	Electronics	Audio	₹ 439.07	142.00
PROD9	Adjustable Dumbbell Set - Hill Plc	Sports	Fitness	₹ 495.72	323.00
PROD10	Ceramic Kitchen Bowl Set - Hamilton Group	Home	Kitchen	₹ 298.86	308.00
PROD11	Hd Smart Tv Stick - Gibson, Hobbs And Weaver	Electronics	Accessories	₹ 449.40	270.80
PROD12	Hair Strengthening Shampoo - Rice Llc	Beauty	Haircare	₹ 115.82	460.00
PROD13	Usb-C Fast Charger - Dawson-Perkins	Electronics	Accessories	₹ 242.23	453.00
PROD14	Decorative Wall Mirror - Luna, Anderson And Bowman	Home	Decor	₹ 48.01	439.00
PROD15	4K Home Projector - Thomas Llc	Electronics	Appliances	₹ 465.34	301.00
PROD16	Beard Grooming Kit - Erickson-Green	Beauty	Grooming	₹ 337.42	256.00
PROD17	Soccer Training Cone Set - Ford, Murphy And Wilson	Sports	Accessories	₹ 144.53	367.00
PROD18	Women'S Denim Jeans - Guerra-Hendrix	Clothing	Bottoms	₹ 245.78	343.00
PROD19	Knee Support Brace - May-Reese	Sports	Equipment	₹ 21.34	65.00
PROD20	Outdoor Sports Water Bottle - Soto-Ellis	Sports	Accessories	₹ 131.51	264.17
PROD21	Multi-Purpose Storage Box - Gardner-Montes	Home	Storage	₹ 123.46	76.00
PROD22	Carbon Fiber Tennis Racket - Flores, Pineda And Rodriguez	Sports	Equipment	₹ 484.62	432.00
PROD23	Men'S Formal Shirt - Moon, Mora And Santos	Clothing	Tops	₹ 248.82	360.00
PROD24	Foldable Camping Chair - Cook Ltd	Sports	Outdoor	₹ 166.06	340.00
PROD25	Women'S Scarf Set - Brown And Sons	Clothing	Accessories	₹ 174.02	268.00
PROD26	Wireless Bluetooth Speaker - Reed-Schmidt	Electronics	Audio	₹ 322.41	213.00
PROD27	Decorative Floor Vase - Khan, Jones And Paul	Home	Decor	₹ 250.02	378.00
PROD28	Sports Stopwatch Timer - Burns-Hansen	Sports	Accessories	₹ 38.76	137.00
PROD29	Fabric Sofa Throw Blanket - Hoffman Group	Home	Decor	₹ 218.48	276.00
PROD30	Stainless Steel Cutlery Set - Stevens, Fernandez And Martin	Home	Kitchen	₹ 19.52	57.00

Column	Solution
Product_Id	Inconsistant value **eg:P-3** to correct form **PRO3**
Product_Brand	Inconsistant value **eg:P-3** to correct form **PRO3**
Product & Brand	cleaned to remove unnecessary spaces and inconsistent formatting **proper(trim(clean(b2)))** **proper(trim(clean(e2)))** then to proceed **MERGE** the columns and the merged column name is **Product_Brand**
Category	Using cleaning process
Sub_Category	Using Cleaning Process
Cost	Missing Values are identified with using the formula **IF(ISBLANK(E2),AVERAGE(E2:E101),E2)**

Stock	Missing Values are identified with using the formula **IF(ISBLANK(F2),AVERAGE(F2:F101),F2)**

And the Non Numeric column are changed to **TEXT** Format

Store_I	Store_Name	Region	City	Store_Type
STORE0	Rivera-Warner	East	Washingtonfurt	Flagship
STORE1	Martinez, Adkins And Johnso	North	Danielmouth	Flagship
STORE2	Black-Horn	South	North Nancyview	Outlet
STORE3	Johnson-Wright	North	Hoodberg	Outlet
STORE4	Alexander Group	South	Port Sharon	Online
STORE5	Gray Ltd	North	New Michaelchester	Outlet
STORE6	Benson-Elliott	East	Port Josephberg	Online
STORE7	Martinez-Mullins	South	Williamsfurt	Online
STORE8	Bean, Jones And Park	West	Thompsonborough	Online
STORE9	Moore, Padilla And Butler	South	East Johnstad	Flagship
STORE10	Hill, Wagner And Ballard	East	East Davidville	Online
STORE11	Aguirre-Keller	South	Lake Michael	Online
STORE12	Thomas, Bridges And Harris	North	Sabrinatown	Online
STORE13	Gardner, Humphrey And Con	East	Johnsonville	Flagship
STORE14	Mcdonald Inc	South	Christopherland	Outlet
STORE15	Shelton Plc	North	East Jenniferstad	Flagship
STORE16	Frederick-Wright	West	New Jimmymouth	Flagship
STORE17	Holloway Ltd	North	Bensonhaven	Flagship
STORE18	Guerrero-Schmitt	East	East Stevechester	Flagship
STORE19	Mann-Austin	East	Vasquezborough	Online

Check all the columns and clean the column with proper,trim and clean

Category	Discount	Unit_Price	Cost	Region	Store_Type	Total_Amount
Sports	0.09	₹ 218.04	245.98	North	Outlet	₹ 217.95
Electronics	0.27	₹ 104.47	403.71	North	Flagship	₹ 312.60
Beauty	0.25	₹ 914.05	43.29	West	Online	₹ 3,655.20
Electronics	0.16	₹ 921.51	392.89	West	Online	₹ 921.35
Beauty	0.18	₹ 745.30	17.31	North	Outlet	₹ 2,235.36
Electronics	0.19	₹ 147.01	465.34	South	Online	₹ 440.46
Home	0.15	₹ 785.31	461.47	South	Online	₹ 1,570.32
Home	0.23	₹ 342.39	138.05	West	Online	₹ 1,368.64
Home	0.07	₹ 448.25	201.41	North	Outlet	₹ 896.36
Home	0.23	₹ 873.17	179.77	East	Flagship	₹ 1,745.88
Sports	0.07	₹ 278.63	166.06	South	Outlet	₹ 278.56
Electronics	0.17	₹ 286.92	326.06	North	Flagship	₹ 1,147.00
Sports	0.01	₹ 644.58	131.24	South	Online	₹ 2,578.28
Electronics	0.30	₹ 316.90	419.28	West	Flagship	₹ 633.20
Sports	0.29	₹ 403.47	166.06	North	Flagship	₹ 1,612.72
Clothing	0.04	₹ 871.34	223.43	South	Flagship	₹ 1,742.60
Sports	0.25	₹ 451.98	240.63	East	Online	₹ 903.46
Home	0.23	₹ 502.69	250.0215714	South	Online	₹ 1,004.92
Electronics	0.25	₹ 559.23	326.06	South	Online	₹ 1,117.96
Beauty	0.04	₹ 513.98	482.75	East	Flagship	₹ 2,055.78
Sports	0.27	₹ 986.04	477.61	North	Flagship	₹ 2,957.31
Electronics	0.22	₹ 589.56	253.33	East	Flagship	₹ 589.34
Clothing	0.03	₹ 979.42	218.22	East	Online	₹ 2,938.17
Beauty	0.08	₹ 719.28	115.82	West	Online	₹ 719.20
Electronics	0.07	₹ 77.60	349.17	East	Online	₹ 232.59
Clothing	0.14	₹ 852.71	261	North	Outlet	₹ 3,410.28
Electronics	0.14	₹ 156.16	431.67	West	Online	₹ 624.08
Electronics	0.15	₹ 815.49	480.9	West	Online	₹ 2,446.03
Electronics	0.15	₹ 74.21	242.23	South	Outlet	₹ 148.12
Home	0.15	₹ 451.67	201.41	North	Outlet	₹ 903.04
Clothing	0.13	₹ 429.38	261	East	Online	₹ 429.25
Clothing	0.01	₹ 148.96	385.56	North	Online	₹ 297.86

Column	Solution
Quantity	The Missing Values Arranged with Use **Median** **MEDIAN(F2:F2000)**
Unit_Price	Using to Find **IF(ISBLANK(O2),AVERAGE(O2:O2000),O2)**
Total amount	These Amount are inconsistent so change with **(O2-N2)*F2**
Discount	**IF(ISBLANK(N2),AVERAGE(N2:N2000),N2)**
Cutomer to sales	**vlookup(c2,cutomer,{3,4,7,8},FALSE)**
Product to sales	**vlookup(p2,product,{2,3,5,6},FALSE)**
Sore to Sales	**vlookup(e2,Store,{3,4,5},FALSE)**

Finally Fetch the All Data in One Sheet Using with **VLOOKUP** and the tables are named so use table name in formula

Fetched the all the datas.

Finally Fetch the All Data in One Sheet Using with **VLOOKUP** and the tables are named so use table name in formula

Cutomer to sales	**vlookup(c2,cutomer,{3,4,7,8),FALSE)**
Product to sales	**vlookup(p2,product,{2,3,5,6},FALSE)**
Sore to Sales	**vlookup(e2,Store,{3,4,5},FALSE)**

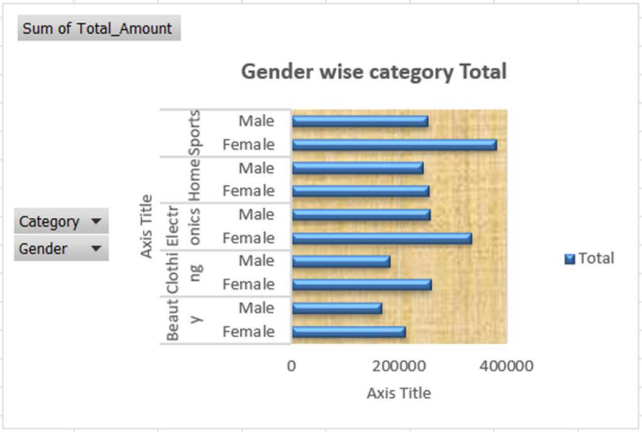
Fetched the all the datas.

Visualization and Insights Arrived with screenshot of visualization Report/Dashboard screenshot:

Sum of Total_Amount	Column Labels			
Row Labels	East	North	South	West
Flagship	357939.1534	377549.41	127402.25	131195.52
Online	374732.2833	138687.69	378571.412	185318.9764
Outlet		243732.8429	239007.2081	
Grand Total	**732671.4367**	**759969.9429**	**744980.8701**	**316514.4964**

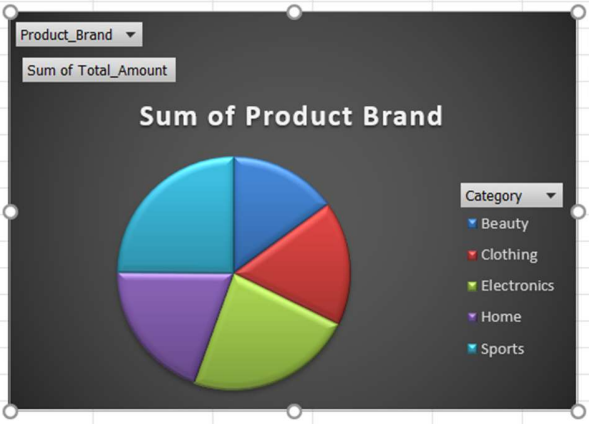


Row Labels	Sum of Total_Amount
Beauty	**381461.2324**
Female	212708.8824
Male	168752.35
Clothing	**442258.9**
Female	259410.76
Male	182848.14
Electronics	**593458.1668**
Female	335606.0504
Male	257852.1164
Home	**501627.1769**
Female	256323.57
Male	245303.6069
Sports	**635331.27**
Female	380920.94
Male	254410.33
Grand Total	**2554136.746**

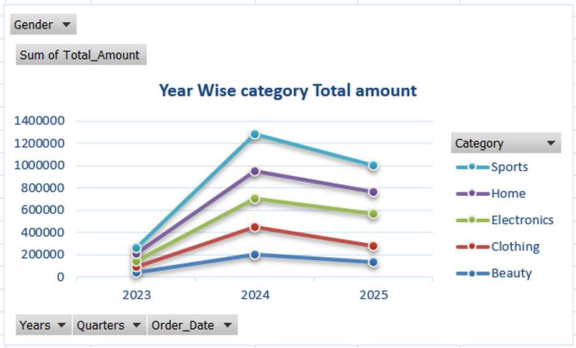


Product_Brand	(All)

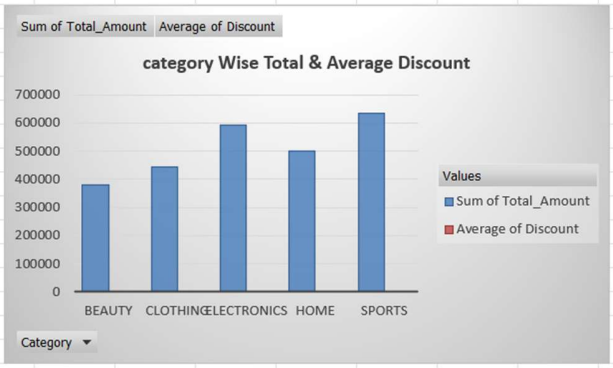
Row Labels	Sum of Total_Amount
Beauty	381461.2324
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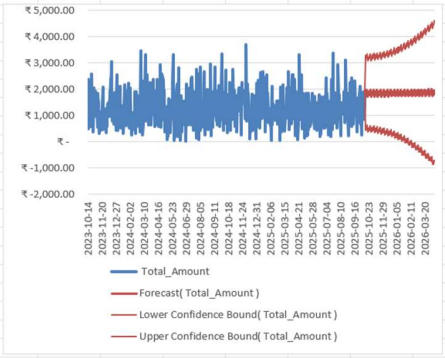
Gender	(All)					
Sum of Total_Amount	Column Labels					
Year	Beauty	Clothing	Electronics	Home	Sports	Sum of Total Amount
2023	43555.57	47156.11	53442.05	63108.63	57259.07	264521.43
2024	205298.1291	244526.87	252244.8963	247705.0729	335626.47	1285401.438
2025	132607.5333	150575.92	287771.2205	190813.474	242445.73	1004213.878
Sum of Total Amount	381461.2324	442258.9	593458.1668	501627.1769	635331.27	2554136.746



Row Labels	Sum of Total_Amount	Average of Discount
Beauty	381461.2324	0.152851974
Clothing	442258.9	0.14488024
Electronics	593458.1668	0.149301364
Home	501627.1769	0.148796423
Sports	635331.27	0.14564257
Grand Total	2554136.746	0.148088176



Order_Date	Total_Amount	Forecast(Total_Amount)	Lower Confidence I	Upper Confidence Boun
2023-10-14	₹ 1,653.26			
2023-10-15	₹ 490.64			
2023-10-16	₹ 2,380.22			
2023-10-17	₹ 1,115.53			
2023-10-18	₹ 771.71			
2023-10-19	₹ 1,755.22			
2023-10-20	₹ 583.87			
2023-10-21	₹ 577.51			
2023-10-22	₹ 2,188.44			
2023-10-23	₹ 627.79			
2023-10-24	₹ 2,574.83			
2023-10-25	₹ 932.94			
2023-10-26	₹ 730.12			
2023-10-27	₹ 552.21			
2023-10-28	₹ 374.29			
2023-10-29	₹ 2,053.50			
2023-10-30	₹ 1,241.48			
2023-10-31	₹ 956.57			
2023-11-01	₹ 1,498.17			
2023-11-02	₹ 1,932.52			
2023-11-03	₹ 1,009.04			
2023-11-04	₹ 1,047.66			
2023-11-05	₹ 1,713.92			
2023-11-06	₹ 1,539.59			
2023-11-07	₹ 1,252.81			
2023-11-08	₹ 539.31			
2023-11-09	₹ 1,844.28			
2023-11-10	₹ 1,271.41			
2023-11-11	₹ 1,535.36			
2023-11-12	₹ 314.19			
2023-11-13	₹ 347.34			
2023-11-14	₹ 1,128.98			



Sales Analysis

What is the total sales amount (sum of Total_Amount)?
What is the total quantity sold?
What is the average transaction value based on Total_Amount?

Product Analysis

Which products have the highest total quantity sold?
Compare total sales amount across product categories.
How do sales amount and quantity sold compare by brand?

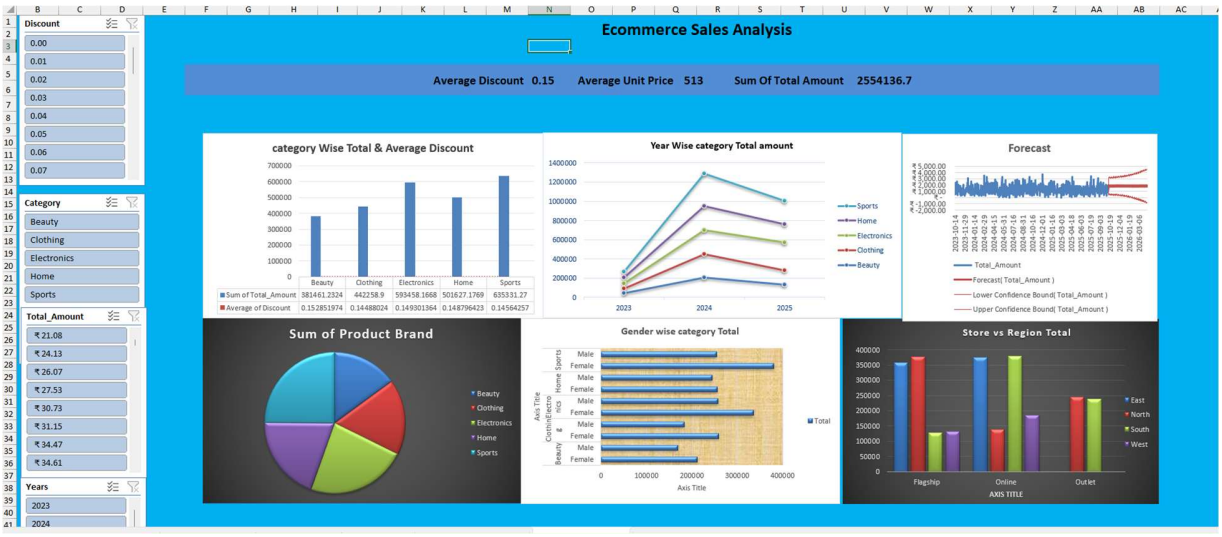
Store Analysis

Which stores generate the highest total sales amount?
Compare total sales amount across regions.
Which cities generate the highest total sales amount?
How does store type (for example, Flagship or Outlet) relate to sales amount?

Customer Analysis

Which customers have the highest total purchase amount?
How does total sales amount vary by customer loyalty level?
Compare total sales amount by customer gender.

Final Dashboard:



Descriptive analysis:

- Sports has the highest total amount among the categories, followed by Electronics and Home.
- Beauty has the lowest total amount compared with the other categories.
- The average discount varies by category, while total sales are much higher for Sports and Electronics.
- 2024 recorded the highest sales amount across all major categories.
- Sales decreased in 2025 compared with 2024, although 2025 still remained higher than 2023.
- Among stores, Online generated strong sales, particularly in the South region.
- The Flagship store performed strongly in the North region.
- The Outlet store generated comparatively lower total sales.
- The gender-wise chart shows differences between male and female sales across categories, with female customers contributing strongly in several categories.

Diagnostic Analysis:

- Sports has the highest sales, suggesting strong customer demand and/or higher transaction values in this category.
- Beauty has comparatively low sales, indicating an opportunity to improve product visibility, promotions, or customer targeting.
- The large increase from 2023 to 2024 suggests that there may have been increased customer demand, successful promotions, expansion of sales channels, or improved product performance.
- The decline from 2024 to 2025 may indicate reduced demand, stronger competition, changes in customer preferences, or less effective promotional activity.
- The strong Online–South combination indicates that the online channel is particularly successful in that region.
- Differences between male and female sales suggest that customer preferences differ by category, which can be useful for targeted marketing.
- Differences between regions and stores indicate that sales performance is not uniform across locations.

Predictive Analysis:

- The forecast indicates the expected future Total Amount based on historical sales trends.
- Historical sales show a strong increase from 2023 to 2024 followed by a decline in 2025.
- If the recent trend continues, future sales may remain relatively stable or experience further pressure rather than repeating the sharp 2024 growth.
- The forecast confidence bounds show that the actual future value could vary from the predicted value.
- Sports, Electronics, and Home should be monitored because they contribute significantly to overall sales.
- Beauty may continue to require attention if its historical performance remains comparatively low.

Prescriptive Analysis:

- The business should focus on its high-performing categories, strengthen the Online channel, and investigate the reasons behind the decline after 2024.
- Targeted promotions should be used to improve low-performing categories such as Beauty.
- Regional and gender-wise customer behavior should be used to develop more focused marketing strategies.
- Inventory and promotional decisions should be continuously adjusted according to forecasted demand.

Conclusion:

The analysis provides a complete view of the business using Descriptive, Diagnostic, Predictive, and Prescriptive Analytics. Descriptive analysis shows that Sports is the leading category and that sales reached their highest level in 2024. Diagnostic analysis indicates that category, region, store type, sales channel, and customer gender contribute to differences in sales performance. Predictive analysis suggests that future sales should be monitored carefully following the decline from the 2024 peak. Prescriptive analysis recommends strengthening high-performing categories and sales channels, improving low-performing categories, focusing on regional and customer-specific marketing, and using forecasts to support inventory and promotional decisions. Overall, the dashboard helps management make data-driven decisions to improve sales and future business performance.